This assignment will not be graded and is only for practice.

Level 1

1) Choose the set of correct options. ***1 point***

- [ ] If v_1 and v_2 are in the kernel of a linear transformation $T : V \to W$, where V and W are two vector spaces, then $v_1 + v_2$ will also be in the kernel of T.
- [ ] Kernel of a linear transformation $T : V \to W$ is a vector subspace of V, where V and W are two vector spaces.
- [ ] If w_1 and w_2 are in the image of a linear transformation $T : V \to W$, where V and W are two vector spaces, then $w_1 + w_2$ will also be in the image of T.
- [ ] Image of a linear transformation $T : V \to W$ is a vector subspace of W, where V and W are two vector spaces.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If v_1 and v_2 are in the kernel of a linear transformation $T : V \to W$, where V and W are two vector spaces, then $v_1 + v_2$ will also be in the kernel of T.
Kernel of a linear transformation $T : V \to W$ is a vector subspace of V, where V and W are two vector spaces.
If w_1 and w_2 are in the image of a linear transformation $T : V \to W$, where V and W are two vector spaces, then $w_1 + w_2$ will also be in the image of T.
Image of a linear transformation $T : V \to W$ is a vector subspace of W, where V and W are two vector spaces.

Consider the following linear transformation:

$$T : \mathbb{R}^3 \to \mathbb{R}^2$$

$$T(x, y, z) = (2x + 3z, 4y + z)$$

Answer questions 2,3,4 and 5, using the information given above.

2) Which of the following matrices corresponds to the given linear transformation T with respect to the standard ordered basis for R^3 and the standard ordered basis for R^2 ? ***1 point***

- ○ $\begin{bmatrix} 2 & 3 & 0 \\ 4 & 1 & 0 \end{bmatrix}$
- ○ $\begin{bmatrix} 2 & 0 \\ 0 & 4 \\ 3 & 1 \end{bmatrix}$
- ○ $\begin{bmatrix} 2 & 0 & 3 \\ 0 & 4 & 1 \end{bmatrix}$
- ○ $\begin{bmatrix} 2 & 4 \\ 3 & 1 \\ 0 & 0 \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\begin{bmatrix} 2 & 0 & 3 \\ 0 & 4 & 1 \end{bmatrix}$

3) Which of the following represents a basis of the kernel of T? ***1 point***

- ○ $\{(-\frac{3}{2}, 0, 1), (0, -\frac{1}{4}, 1)\}$
- ○ $\{(-\frac{3}{2}, -\frac{1}{4}, 1)\}$
- ○ $\{(-\frac{3}{2}, -\frac{1}{4}, 2)\}$
- ○ $\{(2, 0, 3), (0, 4, 1)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(-\frac{3}{2}, -\frac{1}{4}, 1)\}$

4) What will be the dimension of the subspace $Im(T)$?

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

5) Choose the correct option. **1 point**

- ◯ T is an isomorphism.
- ◯ T is one to one but not onto.
- ◯ T is onto but not one to one.
- ◯ T is neither one to one, nor onto.

No, the answer is incorrect.
Score: 0

Accepted Answers:

T is onto but not one to one.

6) Choose the set of correct options. **1 point**

- ☐ There exists a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ whose range is contained in the circle $x^2 + y^2 = 1$.
- ☐ There exists a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ whose range equals the line $x = y$.
- ☐ There exists a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ whose range is contained in the set $S = \{(x, y) \mid x > 0\}$.
- ☐ There exists a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ whose range equals the line $y = \pi x$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

There exists a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ whose range equals the line $x = y$.
There exists a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ whose range equals the line $y = \pi x$.

Level 2

7) Which option represents the kernel and image of the following linear transformation? ***1 point***

$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$
$T(x, y) = (x, 0)$

- ○ $ker(T) = Span\{(1, 0)\}, Im(T) = Span\{(1, 0)\}$
- ○ $ker(T) = Span\{(1, 0)\}, Im(T) = Span\{(0, 1)\}$
- ○ $ker(T) = Span\{(0, 1)\}, Im(T) = Span\{(1, 0)\}$
- ○ $ker(T) = Span\{(0, 1)\}, Im(T) = Span\{(0, 1)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$ker(T) = Span\{(0, 1)\}, Im(T) = Span\{(1, 0)\}$

8) Which of the following linear transformations is an isomorphism between the vector spaces $V = \{(x, y, z) \mid x = y - z, \text{ and } x, y, z \in \mathbb{R}\} \subset \mathbb{R}^3$ and $W = \{(x, y, z) \mid x = y, \text{ and } x, y, z \in \mathbb{R}\} \subset \mathbb{R}^3$? ***1 point***

- ○ $T : V \rightarrow W$, such that $T(x, y, z) = (y, y, 0)$
- ○ $T : V \rightarrow W$, such that $T(x, y, z) = (y, y, z)$
- ○ $T : V \rightarrow W$, such that $T(x, y, z) = (x, y, z)$

- [ ] There does not exist any isomorphism between V and W.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T : V \to W$, such that $T(x, y, z) = (y, y, z)$

9) Which of the following linear transformations is an isomorphism between the vector spaces $V = \{(x, y, z) \mid x = y - z = 0, \text{ and } x, y, z \in \mathbb{R}\} \subset \mathbb{R}^3$ and $W = \{(x, y, z) \mid x = y = 0, \text{ and } x, y, z \in \mathbb{R}\} \subset \mathbb{R}^3$? **1 point**

- [ ] $T : V \to W$, such that $T(x, y, z) = (x, 0, y)$.
- [ ] $T : V \to W$, such that $T(x, y, z) = (0, y, z)$.
- [ ] $T : V \to W$, such that $T(x, y, z) = (0, 0, x)$.
- [ ] There does not exist any isomorphism between V and W.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T : V \to W$, such that $T(x, y, z) = (x, 0, y)$.

10) Consider the linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ such that $T(x, y, z) = (x - y + z, 2x + y - z, -x - 2y + 2z)$. Which of the following options are true about T? **1 point**

- [ ] Range of T is $\{(x, y, z) \in \mathbb{R}^3 \mid x = y + z\}$.
- [ ] Range of T is $\{(x, y, z) \in \mathbb{R}^3 \mid x = 0, \ y + z = 0\}$.
- [ ] Rank of T is 2
- [ ] Rank of T is 1

No, the answer is incorrect.

Score: 0

Accepted Answers:

Range of T is $\{(x, y, z) \in \mathbb{R}^3 \mid x = y + z\}$.
Rank of T is 2

Your score is: 0/10